

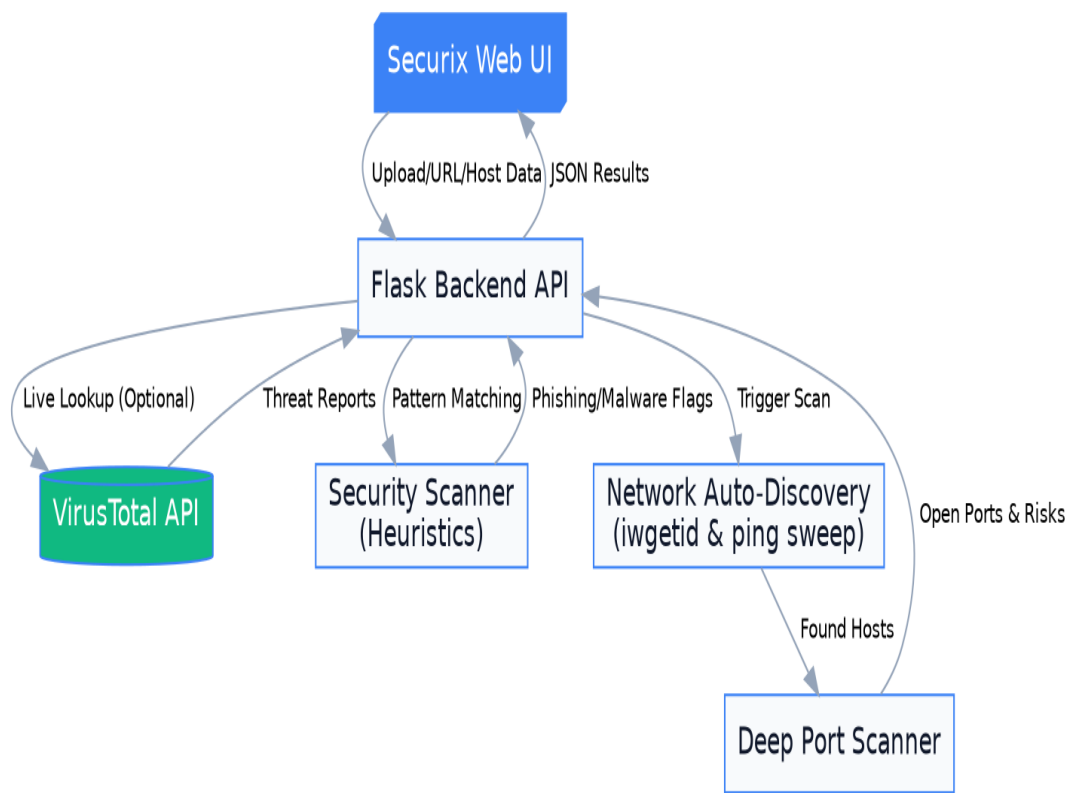
Securix

Advanced Security Diagnostics & Threat Analysis Platform

System Overview

Securix is a comprehensive cybersecurity platform designed to analyze URLs, scan suspicious files, discover network vulnerabilities, and educate users against modern phishing attacks. It combines local heuristic analysis with real-time threat intelligence from VirusTotal.

System Architecture Flow



Core Capabilities

Feature	Description
URL Scanner	Analyzes links for phishing indicators, excessive subdomains, and malicious payloads using I

File Sandbox	Inspects uploaded files for malicious signatures and structural anomalies, supporting live VirusShare integration
Network Discovery	Auto-detects the local Wi-Fi/LAN subnet, probes active hosts, and performs deep port scanning
Email Analyzer	Parses raw email headers to verify SPF, DKIM, and DMARC records to detect spoofing and impersonation
Breach Checker	Cross-references user credentials against HaveIBeenPwned API to identify historical data breaches
Phishing Lab	An interactive mock email client that trains users to spot phishing red flags and social engineering tactics

Security Assurances

The platform is built on a lightweight Flask backend paired with a reactive frontend. API keys and sensitive user configurations are securely stored in a local SQLite database and retrieved dynamically. Threat scanning logic operates securely on the host, ensuring that internal network topology and proprietary files are analyzed safely.